ANDRÉS FELIPE JARAMILLO RUEDA



Backend Engineer (Node.js) | AI Engineer (LLM/Agents) | M.Sc. Industry 4.0

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PROFESSIONAL PROFILE

Backend Engineer with 3.5 years of specialized experience in Node.js / Express and 6 years of overall professional experience in engineering and systems integration. I build and maintain services for a SaaS property management platform serving thousands of Property Management Companies and associations, built on a feature-based modular monolith with Clean Architecture (use cases) supporting MongoDB and SQL Server, with a focus on large-scale document management (AWS S3), cloud integrations (Plaid, Twilio, OpenAI), and cron + AI-driven automation. I lead features end-to-end: analysis, versioned REST API design, implementation, validation, observability, and production data migration with zero downtime.

Hands-on AI Engineering experience: designing, evaluating, and operating production LLM agents with the Strands Agents SDK and AWS Bedrock AgentCore, integrated into multi-tenant NestJS backends for a healthcare-sector application. I implement function-calling/tools, context and session management, RAG and embeddings, plus automated evaluation suites (DeepEval, LLM-as-judge, regression datasets) and guardrails/safety metrics, with attention to per-request cost through prompt caching and token management.

I bring hands-on experience providing support and troubleshooting for React frontend components, along with training in the full MERN stack (React, Context API, Zustand, JWT authentication), which allows me to communicate with strong technical judgment across full-stack teams. Advanced adoption of AI-assisted development with Cursor (agentic workflows, plan-mode, context engineering via rules and skills) to accelerate delivery while maintaining human-in-the-loop code review and testing.

I work under agile methodology with biweekly sprints, code reviews, and PRs via Bitbucket Pipelines. Electronics Engineer (UIS) and M.Sc. in Industry 4.0 (UNIR), a background that provides systems thinking, critical systems integration expertise, and effective communication with technical teams.

WORK EXPERIENCE

AI ENGINEER INTERN | MB Technology Solution LLC | *Jan. 2026 – Present*

Design, evaluation, and operation of production LLM agents (Strands Agents SDK + AWS Bedrock AgentCore), integrated into a multi-tenant NestJS backend for a healthcare-sector application.

- Designed tools/function-calling, context/session management, structured prompting, and hallucination control with the Strands Agents SDK.
- Model invocation via AWS Bedrock and AgentCore, credentials/IAM management, and response streaming handling.
- Responsible for writing and maintaining DeepEval-type evaluation suites, regression datasets (.jsonl), LLM-as-judge, and guardrails/safety metrics.
- Onboarded new use cases to the NestJS backend, respecting the domain layer, with infrastructure service injection (ClaudeApiService, AiImageService).
- Managed prompt caching and input/output/cache tokens, designing features with a focus on cost per request.
- Executed targeted infrastructure-as-code changes with Terraform.
- Developed idiomatic, statically typed Python.
- Implemented and maintained embeddings, retrieval, and basic RAG.

BACKEND DEVELOPER | CINC SYSTEMS COLOMBIA S.A.S | *Mar. 2023 – Jul. 2026*

Development and maintenance of Node.js / Express microservices for the backend of a SaaS property management platform, focused on large-scale document management (file v1/v2, global files, system files on AWS S3), reporting, and multi-tenant data modules (MongoDB + SQL Server), plus mass-migration tooling for legacy client onboarding. Technical coordination with the frontend team on React, including direct involvement resolving incidents and bugs on existing components, providing practical visibility into the frontend ecosystem beyond backend work. Participation in code reviews, versioned REST API design (v1/v2), and biweekly sprint deliveries via Bitbucket Pipelines.

Stack: Node.js 18+ / Express, Mongoose, TypeORM, native MongoDB driver, SQL Server (mssql), AWS SDK v3 (S3, STS, SSO/Identity Center, IAM, Bucket Policies, Lifecycle Rules, presigned URLs, multipart), AWS Amplify Storage (frontend), AWS CLI v2, ExcelJS, express-validator, date-fns-tz, moment-timezone, Sentry, dotenv, readline, image-size, mime-types, Bash, React (support), FSX, ECS.

FSX → S3 Migration — Bulk Migration CLI

Designed and built a Node.js CLI tool (v2.0) used by the operations team to migrate production documents from the legacy WebAxis system (FSX + SQL Server) to the new stack (S3 + MongoDB), preserving SuperAdmin backend models. Processes one client per run with interactive selection, upfront validation, a two-phase pipeline (hierarchical folders → binary files), WebAxis-to-SuperAdmin permission mapping (onlyAdmin/public/committees/userProfiles/allUser), multipart upload for files >100 MB, automatic orphan-folder handling, DRY_RUN mode, and Excel reports with metrics (MB/s, files/hour, error rate). Safe resumption via a numHasMigrated flag in SQL — reruns skip files already migrated or previously failed. Result: a reproducible, auditable pipeline the operations team runs independently of the backend team, with transfer metrics and per-file traceability.

End-to-End S3 Document Management Platform

Built a complete document solution for the platform: backend for the global_file module (CRUD, hierarchy, permissions, cross-DB population between MongoDB ↔ SQL Server, files shared across all PMCs) plus public-facing services (time-limited download URLs, per-user timezone-aware landing pages, date sanitizers). On the infrastructure side, provisioned S3 buckets for dev/uat/prod, defined bucket policies and lifecycle rules, and built a Bash AWS SSO helper (configure-sso.sh) for multi-account authentication with 12-hour sessions. Coordinated with the frontend team, including resolving React component issues, so uploads/downloads run directly against S3 via AWS Amplify, leaving credential issuance and authorization to the backend. Also configured scheduled jobs to generate manifests of all files enabled for AI vectorization. Result: a production-ready, end-to-end solution with S3 costs controlled via lifecycle policies, one-command team onboarding to AWS, and a dataset consumable by the AI pipeline.

Configurable-Filter Reporting

End-to-end module for saved filters and Excel reporting across Charges/Users/Votes/Units, with cascading selectors, multi-tenant support for CINC/ONR (MongoDB and SQL Server), and timezone-aware timeframe handling.

4TH-GENERATION SOLUTIONS ENGINEER | HMV INGENIEROS LTDA | *Jul. 2021 – Mar. 2023*

Designed automation and control architectures for digital substation protection systems; conducted pre-FAT testing on industrial communications using Omicron's Daneo 400 equipment; authored cybersecurity documentation and guidelines per Colombian regulations; designed and integrated HMI systems for digital substation monitoring and operation. Integrated IEC 61850 and IEC 104 protocols with SubNet and designed HMIs using Axon Builder for digital substations. Automated configuration-file generation for industrial communication switches using Python and implemented robust physical industrial communication infrastructure.

CYBERSECURITY ANALYST | IT SECURITY SERVICES | *Feb. 2021 – Jul. 2021*

Created vulnerability alerts for servers and integrated platforms using Python and machine learning development; built dashboards with PCI-DSS indicators.

METROLOGIST IN TRAINING | SOLUMED INGENIERÍA S.A.S | *Feb. 2020 – Aug. 2020 / Sep. 2020 – Dec. 2020*

Performed calibration, verification, adjustment, and qualification of measurement systems in accordance with technical standards. Operated metrology equipment and supported quality assurance in measurement processes.

Participated in expanding the laboratory's scope (length, isothermal media, and spectrometry), a project approved by MinCiencias and INM (Colombia's national metrology institute).

Supported metrological assurance management, root-cause analysis of instrument deviations, and statistical data processing per established methodologies.

EDUCATION

2022 – 2023 | Universidad de la Rioja (UNIR) — **M.Sc. in Industry 4.0** — Logroño, Spain

2014 – 2020 | Universidad Industrial de Santander (UIS) — **Electronics Engineer** — Bucaramanga, Santander

2008 – 2013 | Colegio De Las Américas — **Technical High School Diploma** — Barrancabermeja, Santander

ACHIEVEMENTS

- 2015 — Universidad Industrial de Santander — **Recognition Award**: Co-author of the "Smoking" project, recognized at multiple national and international events.
- 2015 — MILSET Expo-Sciences International, Belgium — **Distinguished Participation**: Represented Colombia at the world youth research fair.
- 2014 — Universidad Industrial de Santander — **Honorable Mention**: For outstanding participation as a speaker at the XVII National Research Meeting, REDCOLSI.

ACADEMIC EXPERIENCE

- 2020 — Universidad Industrial de Santander — Connectivity and Signal Processing Research Group (CPS): **Published a scientific article in the journal Ingeniería y Ciencia (EAFIT)**, "Computational architecture for the inference of a Quantized Convolutional Neural Network for the detection of Atrial Fibrillation," as author. https://doi.org/10.17230/ingciencia.16.32.6
- 2015 — The Jeunesses Scientifiques de Belgique, Brussels, Belgium: **Represented Universidad Industrial de Santander as author of a project at the MILSET Expo-Sciences International**; selected through a series of local, regional, and national events to join Colombia's delegation at the world youth research fair. review
- 2014 – 2016 — Universidad Industrial de Santander — **Control, Electronics, Modeling and Simulation Research Group (CEMOS)**: Research activities as author for the design and development of an electronic device that improves air quality in environments contaminated with tobacco smoke via a high-energy negative-ion generator. Speaker at the local (Barrancabermeja), regional (Ocaña), and national (Tunja) research-seedbed meetings, and at the IDES science, technology and innovation exhibition (Barrancabermeja).

SKILLS

- Node.js
- Express + Mongoose
- NestJS
- MongoDB
- SQL Server
- Python
- AWS (S3, SSO, IAM)
- Docker
- ExcelJS
- Clean Architecture
- React
- JWT
- AI-assisted development (Cursor, Claude)
- Claude Code (agentic workflows)
- MCP (Model Context Protocol)
- Spec-Driven Design
- Skills, Hooks & Custom Commands
- Parallel Agent Orchestration (Git Worktrees)
- Strands Agents SDK
- AWS Bedrock / AgentCore
- RAG, Embeddings & Vector Search
- LLM Evaluation (DeepEval, LLM-as-judge)
- Prompt Engineering & Context Management
- Terraform
- Cybersecurity

CORE COMPETENCIES

- Technical communication with stakeholders
- End-to-end feature leadership
- Code review and mentoring
- Systems thinking

LANGUAGES

- English: Intermediate, in active training
- Portuguese: Intermediate
- Spanish: Native

COURSES & CERTIFICATIONS

- 2026 — Anthropic — **Claude** Code in Action (in progress)
- 2026 — Udemy — **Claude Code:** A Complete Guide for Software Developers (in progress)
- 2026 — Udemy — **React**: From Zero to Expert (knowledge refresh) (in progress)
- 2026 — Udemy — **GoLang**: Language Fundamentals 24h
- 2022 — City University Miami — Continuing Education Certificate in Leadership — 45h
- 2021 — Elastic — **Kibana** Fundamentals — 5h
- 2021 — IT Security Services — **PCI DSS** 3.2.1 Requirements — 8h
- 2020 — SENA — Microsoft **Office** 2016 Tools: Excel — 40h
- 2020 — SENA — Computer-Aided Design – **AutoCAD 2D** — 40h
- 2020 — Academy Professor & STEM Int'l Education — Ladder Logic Programming for Industrial Automation — 4.5h
- 2020 — Google — Digital Skills for Professionals — 40h
- 2016 — Innova Training & Enterprise — Cloud Computing-Based Educational and Business Technologies — 80h

REFERENCES

Available upon request.